

PROGRAMMING FUNDAMENTALS

Spring 2024

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Google Classroom Code

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Identifiers

- Names of things that appear in programs
 - Variables
 - Constants
 - Functions
-
- All identifiers can have letters, digits, and underscore
 - Names must begin with a letter or underscore, cannot have spaces, cannot have + sign, ! Sign etc.
 - Redefined identifiers: cout, cin
 - C++ is a case sensitive language

Using cin & cout

- cin and cout are predefined identifiers
- These are declared in header file iostream within a namespace
- The name of this namespace is **std**
- How to use cin and cout?
 - 1- refer to them as `std::cin` and `std::cout` in your program
 - 2- include the following statement :

`using namespace std;`

This should appear after : `#include <iostream>`

`#include <iostream>`

`using namespace std;`

Data Types

- C++ categorizes data into different types
- Certain operations can be performed on particular types of data

Simple Data Types

- Integral → for number data without a decimal part
- Floating Point → Decimal Numbers
- Enumeration → user-defined data type (chp -7)

To find the number of bytes used

```
cout << endl;  
cout << "A char uses " << sizeof(char) << " byte." << endl;  
cout << "An int uses " << sizeof(int) << " bytes. " << endl;  
cout << "A float uses " << sizeof(float) << " bytes. " << endl;  
cout << "A double uses " << sizeof(double) << " bytes. " << endl;  
cout << endl;
```

Output - ?

A char uses 1 byte.

An int uses 4 bytes.

A float uses 4 bytes.

A double uses 8 bytes.

Q1- Variable name and data type

- i) A variable to store number of students in a class
- ii) Age in years and months
- iii) Number of rooms in your house
- iv) Hourly rate in rupees only
- v) Number of plates bought from a store
- vi) Sales tax in percentage
- vii) Number of characters in a game
- viii) Number of times a student jumped in a contest
- ix) Time to complete exam in minutes and seconds
- x) First Name of a student

Q2:declare and initialize

- i. Month number in a year, value = Feb
- ii. Month number in a year, value = 5
- iii. Registration number of a student in BNU, value=F2023-111
- iv. Name of a city, value = “Lahore”
- v. Number of courses enrolled in Spring2024, value = 6
- vi. Percentage increase in student body at BNU, value = 20.0
- vii. User input to accept ‘Y’ or ‘N’, value = ‘Y’
- viii. Number of balls in a basket, value = 27
- ix. Number of glasses ordered, value = 12
- x. Speed at which you may drive your car, value = 70.25
km/hr

Q3- Let's find the value - i)

```
x = 0;
```

```
y = 5;
```

```
x = x + 1;
```

```
y = y + 1;
```

```
y = y + x;
```

```
cout<<x<<endl;
```

```
cout<<y;
```

ii)

```
i = 10;
```

```
j = 2;
```

```
k = 4;
```

```
j = j * k;
```

```
k = k + j;
```

```
cout<<k;
```

iii)

```
a = 1;
```

```
b = 10;
```

```
c = 100;
```

```
ans = a + b;
```

```
b = b + c;
```

```
ans = b + c;
```

```
ans = ans + 2;
```

```
cout<<ans<<endl;
```

```
cout<<b;
```

iv)

```
int a,b,c;  
a = 10;  
b = a +2;  
c = b * 10;  
a = b + c;  
cout<<c<<endl;  
cout<<a;
```

v)

```
int w, x, y, z;  
x = 1;  
y = x + 2;  
z = y + 10;  
z = x + y + z;  
cout<<z<<endl;  
x = y + z;  
z = x + y + z;  
cout<<z;
```

Q4- Get input-prompt and accept input (i)

- Ask the user to enter his/her Last Name
- Accept the name in a variable after declaring

ii)

- Ask the user to enter distance between his home and university in kilometers
- Accept the distance in a variable after declaring

iii)

- Ask the user to enter his/her score in quiz_1 (out of 25)
- Accept the score in a variable after declaring

iv)

- Ask the user to enter freezing & boiling point of water
- Accept the two separate values with decimal parts in variables after declaring

v)

- Ask the user to enter his aggregate marks in ICT out of 100
- Accept the marks in a variable after declaring

Book Reference

- D.S. Malik (ch - 2)
- Tony Gaddis (ch - 2)